PCBC H Parametrics Practice Test B Name _____ _____/12

Calculator | 30 minutes total for all four questions

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

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Signature _____

SHOW ALL WORK FOR FULL CREDIT!

Practice set: the calculator question comes first, as announced in class. Every question copies a homework problem with new numbers. 3 points each is a practice split of the posted 12.

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandth. *(practice instructions)*

1. The path of a projectile follows the parametric equations

$$x = v_0(\cos\theta)t \qquad \text{and} \qquad y = -16t^2 + v_0(\sin\theta)t + h_0$$

where θ is the angle of inclination of the projectile at the launch, v_0 is the initial velocity of the projectile in feet per second, and h_0 is the initial height of the projectile in feet.

A projectile is launched from the ground with an initial velocity of 96 ft per sec. The parametric equations for its flight are $x = 81.41266t$ and $y = -16t^2 + 50.87224t$.

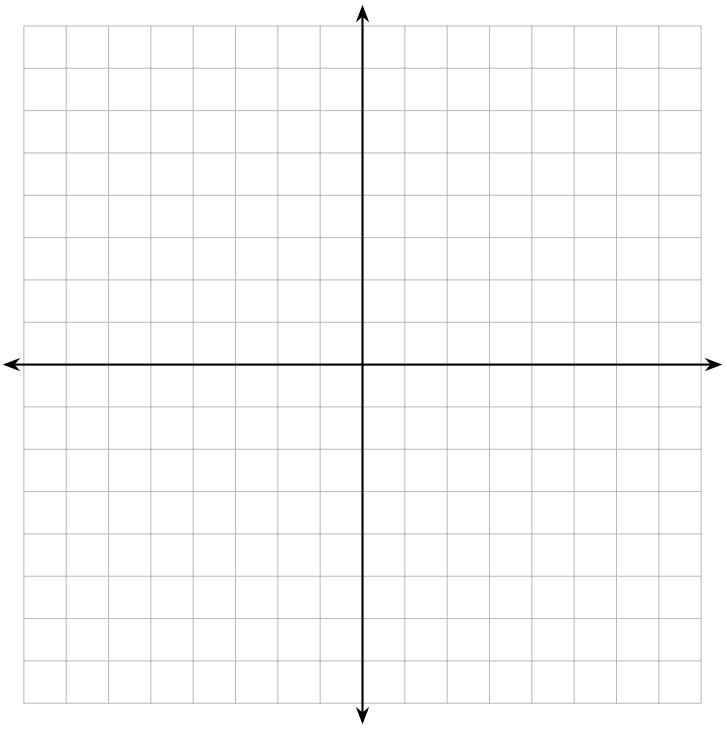
(a) Graph the parametric equations. (b) Approximate θ, the angle the projectile makes with the horizontal at launch. (c) Based on part (b), write parametric equations for the projectile using the cosine and sine functions. (d) For how many seconds is it in flight, and what horizontal distance does it travel? **3 pts**

Non-Calculator 30 minutes total for all four questions

SHOW ALL WORK FOR FULL CREDIT!

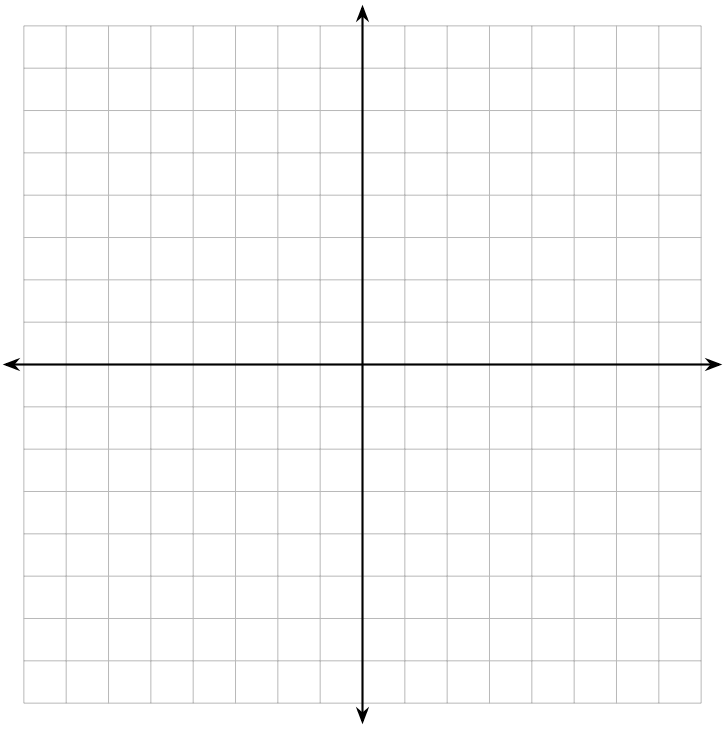
For the plane curve, (a) graph the curve, and (b) find a rectangular equation for the curve. (3 points)

2. $x = 4\sin t, \quad y = 2\csc t, \quad \text{for } t \text{ in } (0, \pi)$ **3 pts**



Find a rectangular equation for the given plane curve, and graph the curve. Show the direction of the curve. (3 points)

3. $x = 2\cot t, \quad y = 4\csc t, \quad \text{for } t \text{ in } (0, \pi)$ **3 pts**



Find two different parametric representations for the equation of the parabola. (3 points)

4. $y = x^2 + 8x + 10$ **3 pts**

Representation 1 *Representation 2*

PCBC H Parametrics Practice Test B: Answer Key

12 points

Each solution follows the method used in class; the gray box shows another way. If the question does not say *simplify*, any form with t gone is fine (a form that shows the graph is best). *Simplify* means get y alone. Direction arrows only when the question asks. The calculator question allows small rounding differences. Every value was checked in Python.

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1. *Calculator* (like homework day 2 #20 and #21)

(a) an arch from $(0,0)$ up to about 40.437 ft and down to $(258.853, 0)$ (b) $\boxed{\theta \approx 32^\circ}$

(c) $\boxed{x = (96\cos 32^\circ)t, \quad y = -16t^2 + (96\sin 32^\circ)t}$ (d) $\boxed{\approx 3.180 \text{ sec}}$, $\boxed{\approx 258.853 \text{ ft}}$

In class	Why
(a) Graphs, Parametric: $x1(t) = 81.41266t$, $y1(t) = -16t^2 + 50.87224t$	Sketch the arch from launch to landing.
(b) $v_0\cos\theta = 81.41266, \quad v_0\sin\theta = 50.87224$	Match to the formula.
$\tan\theta = \dfrac{50.87224}{81.41266}$, so $\theta \approx 32^\circ$	Divide; degree mode for $\tan^{-1}$.
(c) $x = (96\cos 32^\circ)t,\ y = -16t^2 + (96\sin 32^\circ)t$	From the ground, so nothing added.
(d) Trace to $y = 0$; set the trace step to 0.01, then 0.001, until y is as close to 0 as you can get	Precision to the thousandths.
Read $t \approx 3.180$ and $x \approx 258.853$	Same point gives both.

Other way. (b) $v_0 = 96$ is given, so $\cos\theta = \frac{81.41266}{96}$ and $\theta = \cos^{-1}(0.84805) \approx 32^\circ$. (d) Factor: $0 = t(-16t + 50.87224)$, so $t = \frac{50.87224}{16} \approx 3.180$ and $x = 81.41266(3.179515) \approx 258.853$.

Practice scoring (3 pts): 1 for the graph and θ, 1 for the cosine/sine equations, 1 for the time and distance.

Trap: radian mode for $\tan^{-1}$, or using $t = 3.18$ to find x (≈ 258.892).

2. $x = 4\sin t, \quad y = 2\csc t, \quad$ for t in $(0, \pi)$ *Non-Calculator* (like homework day 2 #8)

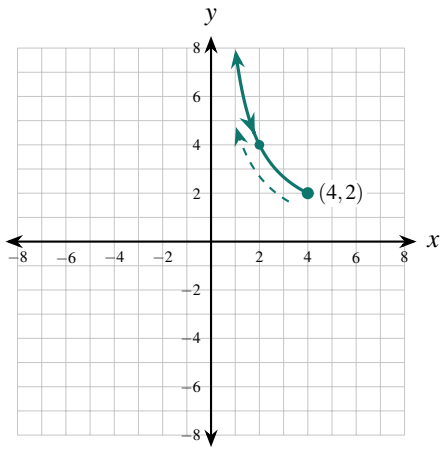
$\boxed{y = \dfrac{8}{x}, \text{ for } x \text{ in } (0,4]}$ $(y = 2 \cdot \frac{4}{x}$ is fine too)

Comes down to $(4,2)$, then goes back up the same path. Arrowhead at the top only.

In class	Why
t-chart: $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$	Parentheses: don't plug in 0 or π.
Plot and connect	Same points twice: it retraces. (Direction isn't asked here.)
$\sin t = \dfrac{x}{4}$	Isolate the trig.
$\csc t = \dfrac{1}{\sin t} = \dfrac{4}{x}$	Flip it.
$y = 2\left(\dfrac{4}{x}\right) = \dfrac{8}{x}$	Substitute into y.
for x in $(0,4]$	x hits 4 at $t = \frac{\pi}{2}$, never 0.

t	(x,y)
$\frac{\pi}{6}$	$(2,4)$
$\frac{\pi}{2}$	$(4,2)$
$\frac{5\pi}{6}$	$(2,4)$

Near $t = 0$ or π, y blows up: arrowhead at the top. $(4,2)$ is reached: it's a turn, not an end.



solid: first pass, coming down
dashed: back up the same path

Other way. Multiply the two equations: $xy = (4\sin t)(2\csc t) = 8$, so $y = \frac{8}{x}$ in one line.

Practice scoring (3 pts): 1 for $y = \frac{8}{x}$, 1 for x in $(0,4]$, 1 for the first-quadrant graph with the turn at $(4,2)$ and the arrowhead at the top.

Trap: $0 < x < 4$. $t = \frac{\pi}{2}$ is allowed, so $x = 4$ is reached.

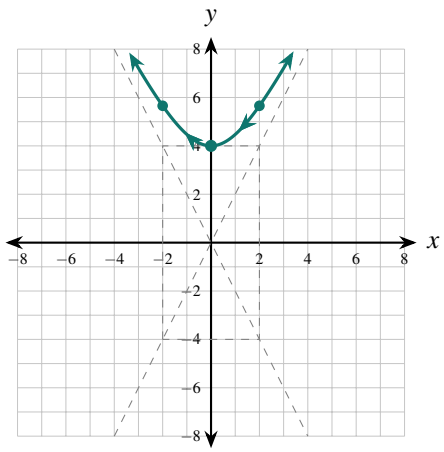
3. $x = 2\cot t, \quad y = 4\csc t, \quad$ for t in $(0, \pi)$ *Non-Calculator* (like homework day 2 #7)

$\boxed{y = \sqrt{4x^2 + 16}}$
Top branch only, traced right to left, arrowheads on both ends.

In class	Why
t-chart: $\frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}$	Parentheses: don't plug in 0 or π.
$\cot t = \frac{x}{2}, \quad \csc t = \frac{y}{4}$	Isolate the trig.
$\csc^2 t - \cot^2 t = 1$, so $\frac{y^2}{16} - \frac{x^2}{4} = 1$	Square and substitute.
Box: up/down 4, left/right 2; dashed diagonals	The asymptotes guide the width.
Connect the points along the asymptotes	Only the top: no negative y in the chart.
$y = \pm\sqrt{16 + 4x^2}$; keep $+$	Solving for y is how you restrict it.

t	(x, y)
$\frac{\pi}{4}$	$(2, 4\sqrt{2})$
$\frac{\pi}{2}$	$(0, 4)$
$\frac{3\pi}{4}$	$(-2, 4\sqrt{2})$

$4\sqrt{2} \approx 5.66$. cot and csc are undefined at 0 and π, so it never stops: arrowheads.



Other way. Pull the 4 out of the root if you want: $y = 2\sqrt{x^2 + 4}$. Same answer; simplifying is optional.

Practice scoring (3 pts): 1 for $y = \sqrt{4x^2 + 16}$, 1 for the top branch only, 1 for the graph with right-to-left arrows.
Trap: stopping at $\frac{y^2}{16} - \frac{x^2}{4} = 1$ (both branches), or writing $y \geq 4$ instead of solving for y.

4. $y = x^2 + 8x + 10$ *Non-Calculator* (like homework day 1 #15 and #16)

$\boxed{x = t,\ y = t^2 + 8t + 10,\ \text{for } t \text{ in } (-\infty, \infty)}$ and $\boxed{x = t - 4,\ y = t^2 - 6,\ \text{for } t \text{ in } (-\infty, \infty)}$

In class	Why
$x = t$, so $y = t^2 + 8t + 10$	Representation 1.
$y = (x^2 + 8x + 16) - 16 + 10 = (x+4)^2 - 6$	Complete the square.
$t = x + 4$, so $x = t - 4$	Let t be the inside.
$y = t^2 - 6$	Representation 2.
for t in $(-\infty, \infty)$ on both	Every x is allowed.

Other way. $x = -4 - t$ also works: $y = (-4-t)^2 + 8(-4-t) + 10 = t^2 - 6$.

Practice scoring (3 pts): 1 for each correct pair, 1 for the t interval on both.

Trap: $t = x + 4$ means $x = t - 4$, not $t + 4$.